

DHRUV PATEL

Aspiring ML/DS Engineer | CS50x | Python, scikit-learn, RAG/LLM Engineering | Open to Internships
+91-9537422869 | dhruv11072005@gmail.com | linkedin.com/in/dhruvpatel0711 |
github.com/DhruvPatel0711 | portfolio

EDUCATION

B.Tech in Computer Science Engineering, Indus University, Ahmedabad June 2027
Current CGPA: 9.1

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java
ML / DS: scikit-learn, pandas, numpy, matplotlib, seaborn, XGBoost, PyTorch
GenAI / LLM: RAG, LangChain, ChromaDB, BGE Embeddings, Prompt Engineering
Web: Next.js, React, Node.js, Express.js, MongoDB, PostgreSQL, Tailwind CSS
Tools: Jupyter Notebook, Git, GitHub, VS Code, Postman, AWS

WORK EXPERIENCE

Full-Stack Development Intern, IITE, Indus University Dec 2024

- Completed full-stack program covering RESTful APIs, modular architecture, and end-to-end deployment using Node.js, Express.js, and MongoDB.

AI Intern, IITE, Indus University June 2024

- Built NLP pipelines using Python and NLTK for text preprocessing, stemming, and lemmatisation; explored LLM integration for conversational AI.

PROJECTS

Laws of Power RAG, Retrieval-Augmented Generation (RAG) Question-Answering System 2026

Python, LangChain (LCEL), ChromaDB, Local Embeddings (BGE), FastAPI, Streamlit

- Built a modular RAG system over a personal 28 document corpus (48 Laws of Power notes): local BGE embeddings (1024 dim), persistent ChromaDB vector store with metadata attached retrieval, a LangChain LCEL orchestration chain with citation constrained grounded generation, and an LLM fallback chain for API reliability; served via a FastAPI backend and Streamlit UI.
- Designed and ran a 60 question retrieval evaluation benchmark across multiple k values (Retrieval@1 68.3% up to Retrieval@10 95.0%), with diagnostics separating *retrieval* failures (answer never reached the LLM) from *generation* failures (correct evidence retrieved, wrong answer generated).

Fix-Risk Predictor, ML System for Commit Level Bug Risk Prediction 2026

Python, Scikit-learn, GitHub REST API, Streamlit

- Self-mined labeled dataset from real GitHub commit history (no public benchmark); built a leakage-safe feature pipeline and compared 6 sklearn models across GroupKFold and time-based validation strategies.
- Deployed as a live inference app on Streamlit Cloud; surfaces fix-risk probability and permutation-importance drivers from a serialized model pipeline, with no live API calls at inference time.

End-to-End ML Pipeline, Multi-Model Classification and Evaluation Framework 2026

Python, scikit-learn, XGBoost, pandas, Matplotlib, Seaborn

- Engineered a complete machine learning pipeline for binary classification, covering data preprocessing, feature scaling (**StandardScaler**), model training, and automated evaluation.
- Benchmarked multiple algorithms, including Logistic Regression, Random Forest, and XGBoost, com-

paring predictive performance across model families.

- Evaluated and visualized model performance across Accuracy, Precision, Recall, and F1-Score using pandas, Matplotlib, and Seaborn.

Odoo POS Cafe (Odoo x Indus University Hackathon 2026, Final Round)

2026

Next.js 14, TypeScript, PostgreSQL, Tailwind CSS

- Offline-capable POS with real-time sync between Cashier terminal and Kitchen Display System; direct PostgreSQL connection, no ORM, for high-performance operation.
- Modules: floor terminal, KDS, admin analytics dashboard.

CERTIFICATIONS & ACHIEVEMENTS

AWS Academy Graduate, Machine Learning for Natural Language Processing Aug 2026

AWS Academy Graduate, Machine Learning Foundations Aug 2026

AWS Academy Graduate, Cloud Foundations Aug 2026

Kaggle Learn, Intro to ML + Pandas + Data Visualisation Kaggle, June 2026

Odoo x Indus University Hackathon 2026, Final Round April 2026

CS50x: Introduction to Computer Science Harvard University / edX, 2025

Artificial Intelligence Internship Program Indus University, June 2024